B.E. 4th Semester Theoretical Examination, 2022-2023

Subject: Computer Organization & Architecture | **Course:** PCC-IT 402

Q1. Address Space & Basics

- **Size of address space:** Address length n bits $\Rightarrow 2^n$ memory locations.
- **Hex representation:** 67 = 43H, 142 = 8EH.
- **IPC:** 35,000 instructions / 17,000 cycles = 2.05.

Q2. Memory Design (2048 Bytes)

- 2048 bytes total. Max ROM = 512 bytes (requires two 256B chips).
- RAM remaining = 1536 bytes (requires six 256B chips).

Q3. Cache Design

- **Write-through vs back:** Through writes to main memory immediately; Back delays write until block eviction.
- **Hit ratio factors:** Cache size, associativity, replacement policy.

Q4. Pipelining Hazards

- **RAW (Read After Write):** Instr 2 reads data produced by Instr 1.
- **WAR (Write After Read):** Instr 2 writes over data before Instr 1 reads it.
- **WAW (Write After Write):** Both write to same register.

Q5. Addressing Modes

- **Immediate:** Operand in instruction. **Direct:** Absolute memory address given. **Indirect:** Address points to pointer.

Q7. DMA Controller

- DMA accesses memory directly bypassing CPU.

- Bus Request (BR) asks CPU for control. Bus Grant (BG) is CPU's acknowledgement.